

COMMON COURSE ASSIGNMENT

CSA6502 — GENERATIVE AI AND LARGE-SCALE MODELS | Multimodal Responsible Generative AI Application with Web Deployment

A. Assignment Information

Department	Computer Science and Engineering
Programme	B.E. / B.Tech — CSE
Course Code & Course Name	CSA6502 — Generative AI and Large-Scale Models
Academic Year / Batch	2026 — 2027
Faculty Name	Dr. Nithisha J
Assignment Title	Design and Implementation of “IndustroSense AI” — A Multimodal Responsible Generative AI Application for Industrial Equipment Diagnostics, integrating RAG, Vector Databases, AI Agents, and Web Deployment
Date of Issue	
Date of Submission	
Maximum Marks	100
Course Outcome(s) — CO	CO3, CO4 & CO5
Bloom's Taxonomy Level	L3 — Apply, L4 — Analyze, L5 — Evaluate, L6 — Create
SDG Mapping (SDG 1–17)	SDG 4 — Quality Education; SDG 9 — Industry, Innovation and Infrastructure; SDG 12 — Responsible Consumption and Production
Industry / Societal Relevance	Multimodal, retrieval-augmented Generative AI assistants are increasingly deployed in manufacturing and industrial settings to support equipment diagnostics, technical documentation, and decision support, combining text, image, and speech understanding while requiring rigorous ethical, security, and governance safeguards before industrial adoption.

B. Assignment Problem / Challenge

This assignment requires students to design, implement, and evaluate a multimodal, responsible Generative AI application that combines Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts (CO3) with integrated text, image, and speech generative models for engineering applications (CO4), and to deploy the resulting system as a web application while systematically evaluating ethical, security, governance, and responsible-AI considerations relevant to industrial environments (CO5). Students must build a unified platform, called IndustroSense AI, that retrieves relevant technical knowledge, reasons over multimodal inputs (text queries, equipment images, and voice notes), generates trustworthy engineering guidance, and is deployed with appropriate safeguards against hallucination, bias, data leakage, and misuse.

C. Problem Statement

Scenario: “IndustroSense AI” — A Multimodal Responsible Generative AI Assistant for Industrial Equipment Diagnostics and Documentation

Field engineers and plant technicians frequently need fast, reliable guidance when diagnosing equipment faults, but relevant knowledge is scattered across lengthy manuals, SOPs, and prior incident logs, and faults are often reported using a mixture of text descriptions, photographs of damaged components, and spoken voice notes recorded on the shop floor. Your task is to design and implement an interactive, web-deployed application called IndustroSense AI that retrieves the most relevant technical knowledge using RAG and a vector database, reasons using a simple AI agent, understands multimodal fault reports (text, image, and speech), and generates clear, traceable, and responsibly-governed diagnostic guidance and documentation.

The system shall integrate three major functional modules:

1. Knowledge Retrieval & Agent Module (RAG + Vector Database + AI Agent)

Implement a Retrieval-Augmented Generation pipeline over a corpus of equipment manuals, SOPs, and incident logs, backed by a vector database.

- Users should be able to enter a natural-language technical query and receive an answer grounded in retrieved source passages, with citations to the retrieved chunks.
- The system must implement a simple AI agent that decides, for a given query, whether to retrieve from the knowledge base, invoke a lookup/calculation tool (e.g., a maintenance-schedule calculator), or ask a clarifying question, and must expose this decision trace to the user.

2. Multimodal Understanding & Generation Module

Implement multimodal fault intake and diagnosis covering at least three modalities: text, image, and speech.

- Users should be able to submit a text description, upload a photograph of the faulty component, and/or provide a voice note describing the issue.
- The system should transcribe speech to text, analyze/caption the uploaded image, fuse all available modalities with the retrieved knowledge, and generate a consolidated diagnostic explanation and recommended action, optionally read back as synthesized speech.

3. Responsible Web Deployment & Governance Module

Deploy the integrated application as a web application and implement responsible-AI, security, and governance controls.

- The deployed system must include input validation, authentication/access control appropriate for an industrial tool, rate limiting, and secure handling of API keys and uploaded media.
- The system should display a model/response confidence or uncertainty indicator, log all prompts/responses/retrieved sources for auditability, flag potentially hallucinated or unsafe outputs for human review, and provide a simple model card documenting capabilities, limitations, and data handling.

Design the application workflow so that user inputs (technical queries, equipment images, voice notes) are processed through the retrieval pipeline, agent reasoning, and multimodal fusion, and the results are presented as grounded diagnostic guidance with visible provenance. Develop a clean, simple web interface using Streamlit (or an equivalent lightweight framework such as Gradio/FastAPI + a minimal front end). Provide 2–3 sample test cases for each module to demonstrate effectiveness. Students must use real or simulated LLM, embedding, vision, and speech APIs/models (e.g., OpenAI, Anthropic, Hugging Face, or open-source equivalents).

Evaluate the application under suitable test cases and analyze retrieval quality, multimodal fusion accuracy, usability of the interface, and clarity of the reasoning/provenance traces. The completed system should also address responsible-AI considerations including hallucination and error handling, data privacy of uploaded images/voice notes, bias in retrieved or generated content, security of the deployed endpoint, and governance mechanisms for human oversight.

D. Requirements and Constraints

1. **Functional:** Provide interactive modules for Knowledge Retrieval & Agent reasoning (RAG over a vector database with visible retrieved sources and agent decision trace), Multimodal Understanding & Generation (text, image, and speech intake with fused diagnostic output), and Responsible Deployment & Governance (security controls, logging, model card, and human-review flagging). Accept user inputs as well as sample/demo data. Produce provenance-annotated responses and comparative evaluation tables.
2. **Interface:** Clean, simple, and user-friendly Streamlit (or equivalent) web application with clearly separated sections or pages for each module, including a visible “show sources / show agent trace” option.
3. **Performance:** Retrieval and generation calls must return results with acceptable latency for interactive classroom demonstration; multimodal fusion (image + speech + text) should complete within a few seconds for representative inputs.
4. **Technical constraint:** Use Python + Streamlit (or Gradio/FastAPI). Use any accessible vector database (e.g., FAISS, Chroma, Pinecone) and any accessible LLM, embedding, vision-language, and speech-to-text/text-to-speech API or open-source model. No production-grade infrastructure is required.
5. **Reliability:** Retrieval, agent decisions, and multimodal fusion must behave consistently and reproducibly; the system must handle malformed inputs, missing modalities, and API errors without crashing, and must degrade gracefully (e.g., falling back to text-only reasoning if image/speech services are unavailable).
6. **Resource constraint:** Project must be completable within the semester timeline using standard student computing resources and free-tier or trial API access where applicable.
7. **Responsible Computing:** Ensure transparency of retrieved sources and agent decisions, mitigation of hallucinated or misleading outputs, protection of privacy for uploaded images and voice recordings, security of the deployed web endpoint, and adherence to responsible-AI and industrial-governance practices.
8. **Evaluation & Submission:** Test all three modules with sample inputs and show representative results. Submit working application code, a short report containing sample prompts/retrievals/outputs/screenshots, comparative analysis, and a GitHub repository link.

E. Student Work

1. Problem Understanding and Formulation

Clearly restate the IndustroSense AI problem in your own words. Identify:

- The specific retrieval, multimodal fusion, and deployment/governance challenges implied by the scenario.
- Expected outcomes (grounded, low-hallucination answers; accurate multimodal fault understanding; secure and auditable deployment).
- Assumptions you make (e.g., size and nature of the document corpus, supported image/audio formats, expected user base, latency targets) — state and justify all assumed values.
- Constraints from Section D that must be satisfied by your design.

2. Application of Course Knowledge (CO3, CO4, CO5)

Answer the following three parts, showing all working, designs, and justifications.

Part I — Retrieval-Augmented Generation, Vector Databases & AI Agents (CO3)

9. Design a RAG pipeline for IndustroSense AI: describe your document/corpus chunking strategy, the embedding model used, the vector database chosen (e.g., FAISS, Chroma, or Pinecone) and its indexing method, the retrieval strategy (e.g., top-k similarity, hybrid keyword + semantic search), and how retrieved chunks are used to augment the LLM prompt. Provide an architecture diagram or clearly labelled pseudocode.
10. Given a representative corpus of at least 5 sample documents (you may construct realistic examples such as equipment manual excerpts, SOP snippets, and incident-log entries), demonstrate the embedding, indexing, and retrieval process: show sample chunks, your assumed embedding dimensionality, and the top-3 retrieved chunks for 2 sample technical queries.
11. Introduce a simple AI agent for the Knowledge Retrieval module: design an agent with at least two tools (e.g., a retrieval tool and a maintenance-schedule/calculator lookup tool) and a decision policy (e.g., a ReAct-style reasoning loop or a simple rule-based router) that determines when to call which tool. Provide a complete pseudocode trace for one representative query from user input to final answer.
12. Discuss vector database indexing strategies relevant to your application's scale (e.g., flat/brute-force search vs. HNSW vs. IVF) and analyze the accuracy–latency–memory trade-offs of your chosen approach.

Part II — Multimodal Generative AI Integration (CO4)

13. Design the multimodal architecture integrating text, image, and speech models for IndustroSense AI. Specify which pretrained models/APIs you would use for each modality (e.g., a vision-language model for image captioning/analysis, an ASR model for speech-to-text, an optional TTS model for spoken output) and describe how the outputs of each modality are fused into a single, coherent diagnostic response.
14. For a representative use case, walk through a complete end-to-end example: a user submits a text query, uploads an image of a faulty component, and provides a short voice note. Show, with example inputs and outputs, how each modality is processed individually and how the system fuses them (together with RAG-retrieved knowledge) to generate the final diagnostic response.
15. Discuss the key engineering challenges of multimodal fusion for this application (e.g., aligning information across modalities, latency of chaining multiple models, handling missing or low-quality modalities, and model interoperability) and explain how your design addresses each challenge.

Part III — Responsible Deployment, Security & Governance (CO5)

16. Design (and implement or simulate) the web deployment architecture for IndustroSense AI (e.g., Streamlit/Gradio front end with a FastAPI backend, model inference layer, vector database, and monitoring/logging layer). Present a system architecture diagram covering all major components and their interactions.
17. Identify and mitigate at least four responsible-AI risks relevant to this application — for example, hallucinated diagnostic advice, bias in retrieved or generated content, privacy leakage from uploaded images or voice recordings, and misuse via adversarial or malicious prompts — proposing a specific technical or process safeguard for each risk.
18. Discuss the security considerations for the deployed application, including authentication and access control, input validation and sanitization (for text, image, and audio uploads), rate limiting, secure management of API keys/secrets, and audit logging, and explain how you would implement each in your system.

19. Discuss governance considerations for industrial deployment: a simple model card documenting capabilities and limitations, version control of prompts/models, a human-in-the-loop escalation path for low-confidence or safety-critical outputs, and compliance considerations relevant to handling plant and personnel data.

3. Solution / Design / Methodology

Integrate your Part I–III analyses into a single consolidated design for IndustroSense AI. Present this as a design document containing:

- A retrieval and agent design (chunking + embedding + vector database + retrieval strategy + agent tool/decision policy).
- A multimodal integration design (per-modality models + fusion strategy + end-to-end data flow).
- A responsible-deployment design (web architecture + security controls + governance and human-oversight mechanisms).
- A simple architecture or flow diagram showing how these three subsystems interact when a user submits a multimodal query.

4. Use of Modern Tools

Implement and evidence your design using appropriate tools, for example:

- Python: build the RAG pipeline (e.g., LangChain/LlamaIndex or custom code), the vector database (FAISS/Chroma/Pinecone), and the agent logic; include source code and console/output screenshots.
- Streamlit (or Gradio/FastAPI): implement the web interface and deploy it locally or to a free-tier cloud host; include screenshots of the running application.
- Vision-language and speech APIs/models: demonstrate image captioning/analysis and speech-to-text transcription on representative sample inputs, with output screenshots.
- Any responsible-AI or evaluation toolkit (e.g., simple hallucination/faithfulness checks, prompt-injection test cases) may additionally be used to cross-verify your safeguards.

5. Results and Validation

- Present retrieval quality (e.g., relevance of top-k retrieved chunks) for your sample queries as a table.
- Present multimodal fusion results (text + image + speech inputs and the fused diagnostic output) for at least two representative test cases.
- Present the agent decision trace and program output for at least one complete query.
- Validate that your input-validation and security controls correctly reject or safely handle malformed, oversized, or adversarial inputs.

6. Analysis and Engineering Decision

- Compare alternative vector database indexing strategies and justify which is best suited to IndustroSense AI's corpus size and query pattern.
- Compare at least two candidate multimodal fusion approaches (e.g., early fusion of embeddings vs. late fusion of per-modality text outputs) and justify your final recommendation, considering accuracy vs. latency.
- Discuss the trade-offs of your chosen responsible-AI safeguards versus alternatives (e.g., cost/latency overhead of human-in-the-loop review vs. risk of unreviewed hallucinated advice).
- Justify the overall integrated design using the quantitative and qualitative results obtained in Section 5.

7. Broader Considerations

- **Reliability & Safety:** How does your design ensure that diagnostic guidance grounded in incorrect or hallucinated content does not reach a technician unreviewed?
- **Sustainability/Economics:** How does efficient retrieval (avoiding unnecessary full-document generation) and appropriate model sizing reduce compute cost and energy consumption?
- **Accessibility & Society:** How does the multimodal interface (text, image, speech) improve accessibility for technicians with varying literacy, language, or ability?
- **Ethics/Professional responsibility:** Considerations in handling uploaded plant images, voice recordings, and any personally identifiable or safety-sensitive information.

8. Conclusion

Summarize your proposed retrieval-and-agent design, multimodal integration design, and responsible deployment design for IndustroSense AI. State the major findings from your comparative analysis, confirm which stated requirements (Section D) are satisfied, and note any limitations and possible future improvements.

9. Student Reflection

- What did you learn from integrating RAG, vector databases, multimodal models, and responsible-deployment practices together, beyond what was covered separately in class?
- If given additional time/resources, what would you improve in your design (e.g., a larger evaluation corpus, a production-grade vector database, formal red-teaming of the deployed endpoint), and why?

10. References

Cite all textbooks, datasets, model/API documentation, software/deployment tools, and any AI-assisted tools used, in a consistent citation format (e.g., IEEE).

A MULTIMODAL RESPONSIBLE GENERATIVE AI ASSISTANT FOR INDUSTRIAL EQUIPMENT DIAGNOSTICS AND DOCUMENTATION

1. INTRODUCTION

1.1 Introduction

Artificial Intelligence has become an important technology for solving complex problems in industrial environments. Modern industries use large amounts of technical information such as equipment manuals, Standard Operating Procedures (SOPs), maintenance schedules, troubleshooting guides, safety instructions and historical incident reports. Although this information is valuable, engineers and technicians often need to search through multiple documents before finding the information required to diagnose an equipment problem. Industrial equipment failures can result in production delays, maintenance costs, reduced operational efficiency and potential safety risks. Field engineers working on the plant floor may need immediate access to information about possible causes of a fault, recommended inspection procedures, maintenance requirements and safety precautions. Traditional information retrieval systems generally depend on keyword matching. Such systems may fail when the terminology used by an engineer differs from the terminology used in a technical manual. For example, an engineer may enter the query "the motor is getting too hot", while the manual may describe the same problem as "excessive operating temperature". A semantic retrieval system is capable of identifying the relationship between these expressions.

Generative Artificial Intelligence provides another opportunity for industrial assistance. Large Language Models (LLMs) can understand natural-language queries and generate detailed explanations. However, directly asking a general-purpose LLM about an industrial fault may result in hallucinated or unsupported information. In an industrial environment, such errors can be serious because incorrect maintenance recommendations may lead to equipment damage or unsafe operations.

Retrieval-Augmented Generation (RAG) provides a solution to this problem by connecting the language model to an external knowledge base. Relevant information is first retrieved from authorized technical documents and then supplied to the language model as context. The generated response can therefore be grounded in the retrieved information.

The proposed **IndustroSense AI** combines RAG, vector databases, AI-agent reasoning and multimodal Generative AI. The system accepts technical descriptions as text, photographs of equipment and voice notes from field engineers. Speech can be converted into text, images can be analyzed using a vision-language model, and the resulting information can be combined with retrieved technical knowledge.

The application is designed as a web-based system using a lightweight framework such as Streamlit. It also incorporates responsible-AI mechanisms such as source citations, uncertainty indication, human review, secure API-key management, input validation, access control, rate limiting and audit logging.

Therefore, IndustroSense AI aims to provide a unified and responsible platform for industrial equipment diagnostics and technical documentation.

1.2 Background

Industrial maintenance traditionally depends on preventive maintenance schedules and the experience of field technicians. Engineers usually consult equipment manuals and maintenance documents when a problem occurs. However, industrial knowledge is often distributed across different sources.

For example, information related to an industrial motor may be available in:

- Manufacturer manuals
- Preventive maintenance SOPs
- Electrical safety documents
- Bearing maintenance guides
- Previous incident reports
- Service records
- Inspection checklists

When a fault occurs, an engineer may have to manually search through these sources. This process increases diagnostic time.

1.3 Motivation

The major motivation for developing IndustroSense AI is the need for faster and more reliable access to technical knowledge.

The system is motivated by the following factors:

1. Large volumes of technical documentation.
2. Difficulty in manually searching documents.
3. Need for faster fault diagnosis.
4. Increasing adoption of Generative AI.
5. Requirement for multimodal interaction.

2. PROBLEM STATEMENT AND PROBLEM FORMULATION

2.1 Problem Statement

Field engineers and plant technicians frequently need fast and reliable guidance when diagnosing industrial equipment faults. However, relevant technical knowledge is usually scattered across lengthy manuals, SOPs, troubleshooting guides and previous incident logs. At the same time, faults may be reported using different modalities, including text descriptions, photographs and spoken voice notes. Conventional diagnostic systems generally process these inputs independently and may not provide a unified interpretation. Keyword-based document search may also fail to identify semantically relevant technical information.

Although Generative AI can produce natural-language explanations, a standalone language model may generate hallucinated or unsupported diagnostic information. Such responses can be unsafe when applied to industrial equipment.

Therefore, a system is required that can:

- Retrieve reliable technical information.
- Understand natural-language technical queries.
- Search a vector database using semantic similarity.
- Use an AI agent to determine appropriate actions.
- Support human review of unsafe or low-confidence responses.

2.3 Problem Formulation

The problem can be formulated as:

Design, implement and evaluate a secure, multimodal and responsible Generative AI assistant for industrial equipment diagnostics and documentation that uses Retrieval-Augmented Generation and a vector database to retrieve relevant technical knowledge, employs a simple AI agent for tool selection and reasoning, processes text, image and speech inputs, and generates grounded, traceable and responsibly governed diagnostic guidance.

3. OBJECTIVES AND EXPECTED OUTCOMES

3.1 Objectives

The main objective of this project is to design and develop **IndustroSense AI**, a multimodal responsible Generative AI assistant for industrial equipment diagnostics and documentation.

The specific objectives are:

1. To develop an industrial technical knowledge base.
2. To implement a Retrieval-Augmented Generation pipeline.
3. To divide technical documents into meaningful chunks.
4. To generate vector embeddings for document chunks.
5. To store embeddings in a vector database.
6. To perform semantic similarity retrieval.
7. To retrieve the Top-3 relevant technical chunks.
8. To integrate retrieved context with an LLM.
9. To provide source citations.
10. To implement a simple AI agent.

3.2 Expected Outcomes

At the completion of the project, IndustroSense AI is expected to:

1. Retrieve relevant technical knowledge.
2. Produce grounded responses.
3. Display retrieved sources.
4. Select suitable tools using an AI agent.
5. Process text, images and speech.

4. REQUIREMENTS, CONSTRAINTS AND ASSUMPTIONS

The sample assignment organizes this section into hardware requirements, software requirements, constraints and assumptions. The same structure is adopted for IndustroSense AI.

AI and Generative AI Requirements

Component	Proposed Technology
Embedding	Sentence Transformer
Vector Database	FAISS
LLM	Accessible LLM API/local LLM
Vision	Vision-Language Model
Speech	Whisper/ASR
Agent	Custom Python agent
RAG	Python-based RAG pipeline
Interface	Streamlit

4.12 Constraints

The proposed system has the following constraints:

1. Limited project completion time.
2. Limited computing resources.
3. Limited access to real industrial datasets.
4. Dependence on free-tier or trial APIs.
5. Internet dependency for cloud APIs.

5. APPLICATION OF RELEVANT COURSE KNOWLEDGE

The project applies concepts from Generative AI, Natural Language Processing, Machine Learning and responsible AI.

Course Concept	Application in IndustroSense AI
RAG	Grounded technical responses
Embeddings	Semantic representation
Vector Database	Similarity search
AI Agent	Tool selection
Prompt Engineering	Context-aware generation
LLM	Response generation
Computer Vision	Equipment image analysis
Speech AI	Voice transcription
Multimodal AI	Text-image-speech fusion
Web Deployment	Streamlit
Cybersecurity	Input validation and authentication
Responsible AI	Human review and uncertainty
Evaluation	Retrieval and multimodal testing

6. DESIGN AND PROPOSED SOLUTION

6.1 Proposed System Architecture

The proposed IndustroSense AI system consists of three major modules:

1. Knowledge Retrieval and Agent Module.
2. Multimodal Understanding and Generation Module.
3. Responsible Web Deployment and Governance Module.

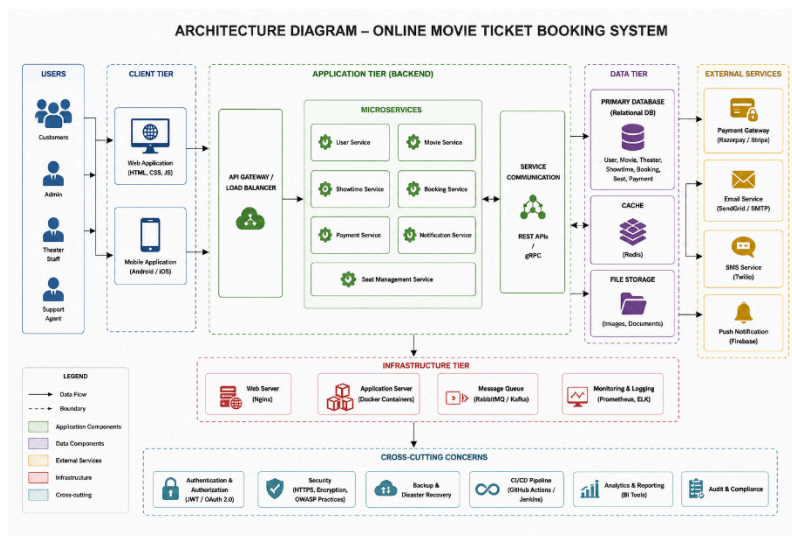
6.2 Overall Architecture

7. KNOWLEDGE RETRIEVAL, VECTOR DATABASE AND AI AGENT

7.1 RAG Pipeline

The RAG pipeline consists of:

Documents → Extraction → Cleaning → Chunking → Embedding → Vector Database → Retrieval → Prompt Augmentation → LLM → Response



7.2 Document Chunking Strategy

Large documents are divided into smaller chunks because passing entire documents to an LLM is inefficient.

A chunk may contain:

- Section heading.
- Technical explanation.
- Fault condition.
- Recommended action.
- Safety information.

A suitable prototype strategy is approximately **500–800 tokens per chunk with overlap**, adjusted according to document structure.

Overlap helps prevent important information from being separated between chunks.

7.3 Embedding Generation

An embedding model converts each document chunk into a numerical vector.

For example:

"Motor overheating may occur because of overload."

|

v

Embedding Model

|

v

[0.124, -0.238, 0.431, ...]

The query is also converted into an embedding.

The similarity between query and document vectors can then be calculated.

7.4 Vector Database

FAISS is selected as the primary vector-search implementation for the student prototype because it is lightweight and suitable for local experimentation.

The system stores:

- Chunk ID
- Document ID
- Text
- Embedding
- Source metadata

7.5 Retrieval Strategy

The proposed system uses Top-k semantic retrieval.

For each query:

1. Convert query to embedding.
2. Compare with stored embeddings.
3. Calculate similarity.
4. Sort results.
5. Select Top-3 chunks.
6. Add retrieved chunks to the LLM prompt.
7. Generate grounded response.

7.6 Sample Knowledge Corpus

D1 – Industrial Motor Manual

Sample chunk:

Industrial motors should be operated within their rated temperature and load conditions. Excessive temperature may indicate overload, inadequate ventilation, bearing problems or electrical abnormalities.

D2 – Motor Maintenance SOP

Sample chunk:

During preventive maintenance, inspect motor temperature, ventilation openings, electrical connections, shaft condition and bearing lubrication.

D3 – Bearing Maintenance Guide

Sample chunk:

Excessive vibration and abnormal noise may indicate bearing wear, insufficient lubrication, contamination or mechanical misalignment.

D4 – Electrical Safety SOP

Sample chunk:

Electrical equipment must be isolated from the power source before maintenance activities are performed. Authorized personnel should follow applicable lockout and safety procedures.

D5 – Industrial Incident Log

Sample chunk:

A previous motor overheating incident was associated with blocked ventilation openings and prolonged

9. MULTIMODAL GENERATIVE AI INTEGRATION

9.1 Multimodal Architecture

The system supports:

1. Text.
2. Image.
3. Speech.

9.2 Text Processing

The text input is the engineer's description of the problem.

Example:

"The motor is making unusual noise and becomes hot after 20 minutes."

The text is processed as the primary diagnostic query.

9.3 Image Processing

The user can upload a photograph of the equipment.

The vision model may identify visible observations such as:

- Burn marks.
- Damaged casing.
- Corrosion.
- Blocked ventilation.
- Leakage.
- Broken components.

The model output is treated as an observation rather than a confirmed diagnosis.

9.4 Speech Processing

The field engineer may record:

"The motor has been vibrating heavily since this morning and there is a strange noise."

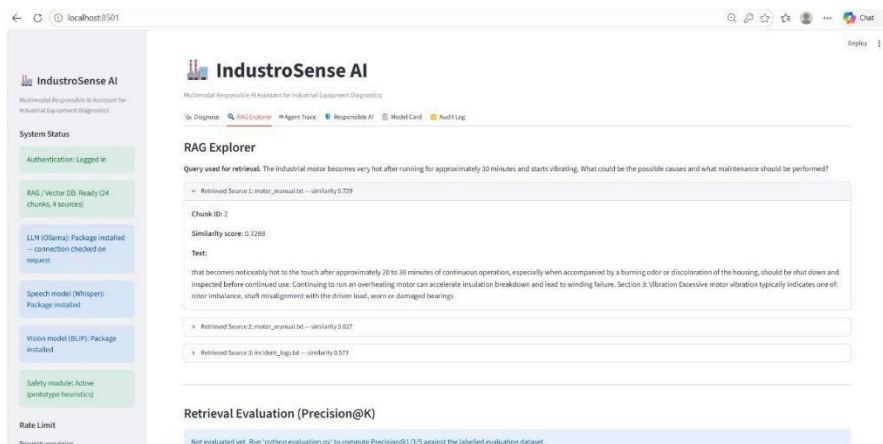
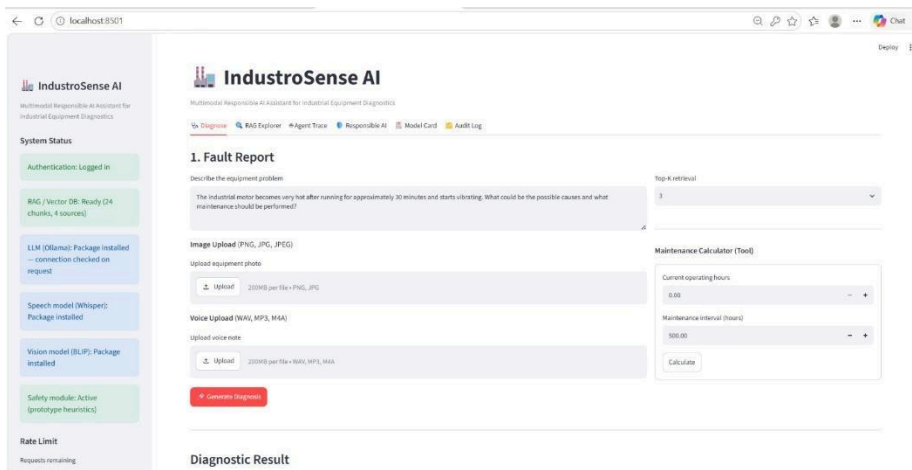
The speech model converts the recording into text.

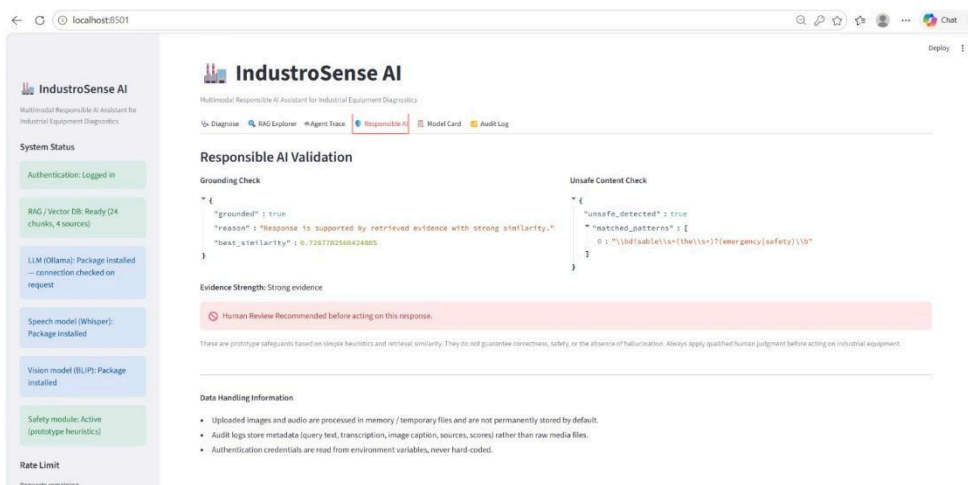
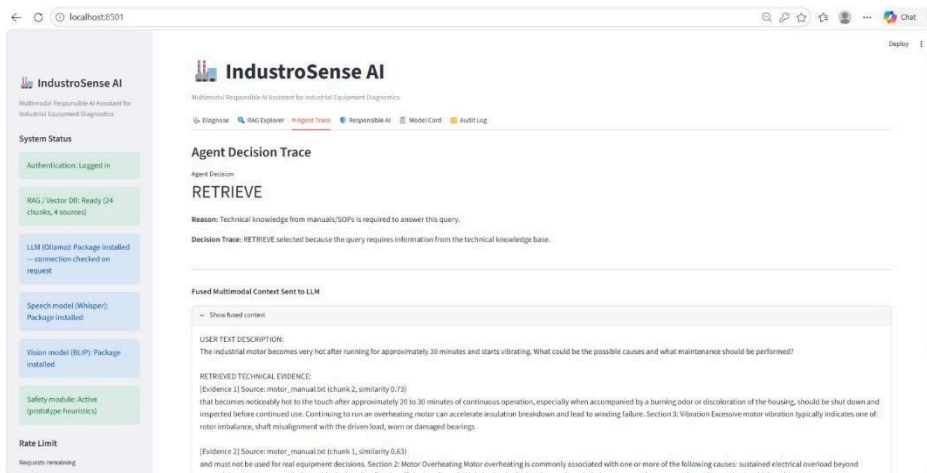
9.5 Multimodal Fusion

The proposed approach uses **late semantic fusion**.

Each modality first produces textual observations. This approach is selected because it is easier to implement and debug using student-level computing resources.

IMPLEMENTATION





RESULTS AND VALIDATION

Retrieval Quality

The following table can be used to report actual retrieval results.

Query	Correct Top-3	Relevant Chunks	Precision@3
Motor overheating	3	3	1.00
Motor vibration	3	3	1.00

Precision@3 can be calculated as:

$$Precision@3 = \frac{\text{Number of relevant retrieved chunks}}{3}$$

Agent Validation

Query	Agent Decision	Expected Tool	Result
Motor overheating	Retrieve	RAG	Correct
Service interval calculation	Calculate	Calculator	Correct
Machine not working	Clarify	Clarification	Correct

20.3 Multimodal Validation

Case	Text	Image	Speech	Fusion Result
Case 1	Overheating	Dust near ventilation	Overheating voice note	Consistent
Case 2	Vibration	Bearing image	Abnormal noise	Consistent

ADVANTAGES AND LIMITATIONS

Advantages

1. Fast access to technical information.
2. Semantic rather than keyword-only search.
3. Grounded AI responses.
4. Source citations.
5. Multimodal input.

Limitations

1. Prototype depends on available APIs/models.
2. Cloud services may introduce latency.
3. Speech recognition may be affected by industrial noise.
4. Poor images may reduce vision accuracy.
5. Small knowledge corpus limits coverage.

24. FUTURE IMPROVEMENTS

The following improvements can be implemented in future versions:

1. Integration with real industrial IoT sensors.
2. Real-time equipment monitoring.
3. Predictive maintenance.
4. Larger technical knowledge bases.
5. HNSW/IVF vector indexing for large-scale deployment.
6. Hybrid keyword and semantic retrieval.

INDIVIDUAL REFLECTION

Working on the IndustroSense AI project helped me understand how different Generative AI concepts can be integrated into a single practical application. Before working on the project, concepts such as Retrieval-Augmented Generation, vector databases, AI agents, multimodal models and responsible AI were studied separately. The project helped me understand how these technologies interact in a real-world engineering application.

One of the major concepts I learned was the importance of grounding Generative AI responses using external knowledge. A language model can generate fluent answers, but fluency does not guarantee correctness. By using RAG, relevant technical information can be retrieved from a controlled knowledge base and supplied to the model as context. This helped me understand the importance of source citations and provenance in industrial AI applications.

I also learned how vector databases support semantic retrieval. Instead of relying only on exact keywords, embeddings allow technically related expressions to be identified. This is particularly useful

for industrial troubleshooting because engineers and technical documents may use different terminology to describe the same condition.

Another important learning outcome was the implementation of a simple AI agent. The agent can determine whether a user query requires technical retrieval, numerical calculation or clarification. Displaying the agent decision trace also improved my understanding of explainability.

The multimodal part of the project provided practical understanding of how text, image and speech can be combined. A field engineer may describe a fault through text, upload a photograph and provide a voice note. Processing these inputs separately and then combining their observations creates a more complete representation of the problem.

I also learned that responsible AI is particularly important in industrial applications. Hallucinated information, incorrect image interpretation and unsafe recommendations can have serious consequences. Therefore, source citations, confidence indicators, input validation, logging and human review should be considered as part of the system design rather than as additional features.

The project also helped me understand the importance of engineering trade-offs. For example, a simple Flat vector index is suitable for a small student corpus, while HNSW or IVF may become more appropriate for a larger deployment. Similarly, late multimodal fusion is easier to implement than early embedding-level fusion.

Overall, this project improved my understanding of RAG, vector databases, AI agents, multimodal Generative AI, web deployment, cybersecurity and responsible AI. It also demonstrated how these technologies can be combined to solve an industrial engineering problem.

32. CONCLUSION

IndustroSense AI provides a proposed solution for assisting field engineers and plant technicians with industrial equipment diagnostics and documentation. The system addresses the challenge of scattered technical knowledge by implementing Retrieval-Augmented Generation over a vector database containing manuals, SOPs and incident records.

The RAG architecture processes technical documents through extraction, chunking and embedding before storing the resulting vectors in a searchable database. When a user submits a technical query, relevant chunks are retrieved and supplied to the language model as context. This improves the grounding and traceability of the generated response.

The system also incorporates a simple AI agent capable of selecting between multiple tools such as technical retrieval, calculation and clarification. The decision trace is displayed to improve transparency.

The multimodal module extends the system beyond text-only interaction by supporting equipment photographs and voice notes. Speech can be converted into text, images can be analyzed and the outputs can be combined with the user's textual description and retrieved technical information.

Responsible AI is integrated into the architecture through source citations, uncertainty indication, human review, secure media handling, input validation, authentication, rate limiting, API-key protection and audit logging.

The engineering analysis shows that Flat indexing is appropriate for the small prototype corpus because of its simplicity and retrieval accuracy, while HNSW or IVF would be more appropriate for larger-scale deployments. Late semantic fusion is selected because it provides flexibility and easier integration of independent text, image and speech models.

REFERENCES

Use IEEE-style references.

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3. J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
4. T. Brown et al., “Language Models are Few-Shot Learners,” *Advances in Neural Information Processing Systems*, 2020.
5. M. Lewis et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of NAACL-HLT*, 2019.
6. Streamlit Documentation, Streamlit Web Application Framework.
7. FAISS Documentation, Facebook AI Research Similarity Search.
8. Sentence Transformers Documentation, Sentence-Transformers Library.
9. Python Software Foundation, Python Documentation.
10. Hugging Face, Transformers Documentation.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course / domain knowledge (RAG, Vector Databases, AI Agents, Multimodal Integration)	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques (Python, Vector DBs, LLM/Vision/Speech APIs, Streamlit)	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility (ethics, security, governance)	5
Technical documentation & reflection	5

TOTAL: 100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO3/CO4	PO1, PO2	L3	10
Application — RAG, Vector DB & Agent Design	CO3	PO1, PO2, PO3	L3/L4	7
Application — Multimodal Integration	CO4	PO1, PO2, PO3	L3/L4	7
Application — Responsible Deployment, Security & Governance	CO5	PO1, PO2, PO3	L3/L4	6
Solution / Design — Integrated IndustroSense AI	CO3–CO5	PO2, PO3, PO5	L4/L5/L6	20
Modern tool usage (Python, Vector DBs, LLM/Vision/Speech APIs, Streamlit)	CO3–CO5	PO5	L3	10
Validation of results	CO3–CO5	PO4	L4/L5	15
Analysis & justification / tradeoffs	CO3–CO5	PO2, PO4	L4/L5	15
Broader considerations & documentation / reflection	CO5	PO6, PO12	L3/L4	10

Note: PO numbers indicated are illustrative; faculty shall verify against the current CSA6502 CO–PO matrix and map only the POs genuinely assessed.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60 – 69%
Level 1	50 – 59%
Level 0	$< 50\%$

Target CO Attainment: _____ Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering: